Here north west corner cell is (3,C) Allocate as much as possible in the first cell so that either the capacity of third row is exhausted or the destination requirement of the third column is exhausted.

i.e. x_{33} = min (4,18) = 4

	C	a_i
3	(4) 7	4/0
4	2	14
b_j	18/14	

Reduced transportation table and final allocation is $x_{44} = 14$

	C	a_i
4	(14) 2	14/0
b_j	14/0	

Thus we have the following allocations

	A	B	C	a_i
1	(5) 2	7	4	5
2	(2) 3	(6) 3	1	8
3	5	(3) 4	(4) 7	7
4	1	6	(14) 2	14
b_j	7	9	18	

Transportation schedule : 1→ A, 2→A, 2→B, 3→B, 3→C, 4→C

The total transportation cost.

$= (5\times2) + (2\times3) + (6\times3) + (3\times4) + (4\times7) + (14\times2)$

= ₹ 102

Example 10.2

Determine an initial basic feasible solution to the following transportation problem using North West corner rule.

	D_1	D_2	D_3	D_4	Availability
O_1	6	4	1	5	14
O_2	8	9	2	7	16
O_3	4	3	6	2	5
Require-ment	6	10	15	4	35

Here O_i and D_j represent i^{th} origin and j^{th} destination.

Solution :

Given transportation table is

	D_1	D_2	D_3	D_4	Availability (a_i)
O_1	6	4	1	5	14
O_2	8	9	2	7	16
O_3	4	3	6	2	5
Requirement (b_j)	6	10	15	4	35

Total Availability = Total Requirement. Therefore the given problem is balanced transportation problem.

Hence there exists a feasible solution to the given problem.

First allocation:

	D_1	D_2	D_3	D_4	a_i
O_1	(6) 6	4	1	5	14/8
O_2	8	9	2	7	16
O_3	4	3	6	2	5
b_j	6/0	10	15	4	35

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